CS101- Introduction to Computing and Artificial Intelligence (2 CH)	CS/AI/DS/SE/CYS/CE

Pre-Requisite: Nil

Instructor: **Muhammad Qasim Riaz**
T/A: **Engr. Amna Arooj**
Office # CS-20, Third Floor, New Academic Block, GIK Institute, Ext. 2172
Email: qasim.riaz*@giki.edu.pk*
Office Hours: 3:00 pm - 5:00 pm (Monday/Wednesday)

Course Introduction

This course focuses on a breadth-first coverage of computer science and Artificial Intelligence (AI) disciplines. Broadly, the course covers contents, like introducing computing environments, general application software, basic computing hardware, operating systems, desktop publishing, Internet, software applications and tools and computer usage concepts, supervised and unsupervised learning, text processing and AI applications; introducing Software engineering and Information technology within the broader domain of computing, C/C++ Language

Course Contents

1. **Basics of Computer Software, Hardware, and AI**
   Computers & Applications, Introduction to Hardware and Software categories, Data Representation, Number Systems, Conversion Methods, ASCII / Unicode, Microprocessors, Memory, Storage Devices, AI learning techniques, AI applications.
2. **Basic Computer Programming**
   Algorithms, Flowcharts & Pseudocode, Assignment Operators, If Selection Statement, If… Else Selection Statement, Nested Control Structures, switch Multiple-Selection Statement, Passing Arrays to Functions, Searching Arrays, Pointers, Library Functions and Header Files

Mapping of CLOs and PLOs

Sr. No	Course Learning Outcomes[+]	PLOs[*]	Blooms Taxonomy
CLO_1	Know the primary techniques computers used to learn patterns for the AI-based autonomous systems.	PLO 1	C1 (Knowledge)
CLO_2	Apply fundamental concepts of computer programming using mathematical concepts and operations.	PLO 1	C3 (Application)
CLO_3	Apply functions for simple modular program with elementary constructs of a programming language.	PLO 1	C3 (Application)
	[+]Please add the prefix "Upon successful completion of this course, the student will be able to"		

CLO Assessment Mechanism

Assessment tools	CLO_1	CLO_2	CLO_3
Quizzes	100%	50%	
Assignments		50%	50%
Project			50%
Midterm Exam	50%	50%	
Final Exam		50%	50%

Overall Grading Policy

Assessment Items	Percentage
Quizzes	10%
Assignments + Project	10% + 10% = 20%
Midterm Exam	30%
Final Exam	40%

Text and Reference Books

Text books:

- Paul J. Deitel and Harvey M. Dietel, " C++20 for Programmers ", 9th Edition, Deitel & Associates, Inc. (2022)

Reference books:

- Chakraborty, U., Banerjee, A., Saha, J. K., Sarkar, N., & Chakraborty, C. (Eds.). (2022). Artificial intelligence and the fourth industrial revolution. CRC Press.
- Parsons, J. J. (2022). New Perspectives Computer Concepts Comprehensive. Cengage Learning.

Administrative Instruction

- According to institute policy, "Students are required to attend 100% classes of courses registered and cannot abstain from even a single class. For circumstances beyond their control, students must apply for leave on the prescribed leave application form".
- Assignments must be submitted as per instructions mentioned in the assignments.
- In any case, there will be no retake of (scheduled/surprise) quizzes.
- For queries, kindly follow the office hours in order to avoid any inconvenience.

Computer Usage

- Most of the assignments must be solved through computers using VS Code with Min GW

Lecture Breakdown

- Lecture 01. Modern Computer, Data Representation in Computer
- Lecture 02. Data Representation in Computer
- Lecture 03. Artificial intelligence and the computer learning, Software and hardware for AI-based systems
- Lecture 04. Basics of natural language processing and its applications
- Lecture 05. Algorithms, Flowcharts, Pseudocode, Memory Concepts, Athematic operators
- Lecture 06. Types of Programming Languages (for desktop, web, mobile, and others)
- Lecture 07. Variables in C++ and basis I/O operations
- Lecture 08. Programming logic development and case studies
- Lecture 09. C++ control structuresecture 09. Control Structures: If Selection Statement
- Lecture 10. Control Structures: If……..Else Selection Statement
- Lecture 11. Nested Control Structures
- Lecture 12. Switch Multiple Selection Statement, Increment Decrement Operator
- Lecture 13. The Essential of Repetition: Counter-Controlled Repetition
- Lecture 14. The Essential of Repetition: Sentinel-Control Repetition
- Lecture 15.While Repetition Statement, do….while Repetition Statement
- Lecture 16. For Repetition Statement
- Lecture 17. Break and Continue Statement
- Lecture 18. Functions: Definition, Prototypes, Calling
- Lecture 19. Call by Value
- Lecture 20. Call by Reference
- Lecture 21. Random Number Generation
- Lecture 22. Storage Classes, Scope Rules
- Lecture23. Arrays Declaration
- Lecture 24. Arrays Examples
- Lecture 25. Passing Arrays to Functions
- Lecture 26. Searching Arrays
- Lecture 27. Pointers
- Lecture 28.Passing Pointers as Function Parameters
- Lecture 29. Library Functions
- Lecture 30. Header Files